

StomaML: Computer Vision for Stomatal Density Detection in Birch Leaves

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Introduction

Stomatamal densities fluctuate by environmental factors and analysing this is crucial for understanding plant responses to environmental and climate changes. Currently, the stomata for birch tree leaves must counted manually when studying hundreds of samples, which is extremely time consuming and forms a significant bottleneck. This project addresses the need for automated stomata detection and density calculation in birch tree leaves, enabling large-scale genomic studies. With automation the analysis of 200+ samples could be done quickly, and a subsequent correlation study would be made significantly easier.

We have developed a standalone web application that automatically gives detects and quantifies stomata in images of birch leaves. Users will upload batches of leaf images, and the system will output a CSV file containing stomata counts per image, countable area of stomata (excluding veins and artifacts, stomatal density & stomata size measurements. The success criterion will be accurately identifying the stomata, mask veins and artifacts and calculate density values that on average closely match manual counts.

Data

We used 376 microscope images that were magnified 10-fold. The images had been taken and were provided by the customer.

Each image had an average of 40 stomatal pores. In addition, there are trichomes, veins, blurred areas as well as what we call *nothing* —or areas outside of the leaves or small holes. Please refer to Figure 1 for an example.

Methods

We used Roboflow for annotating and training our computer vision models [1].

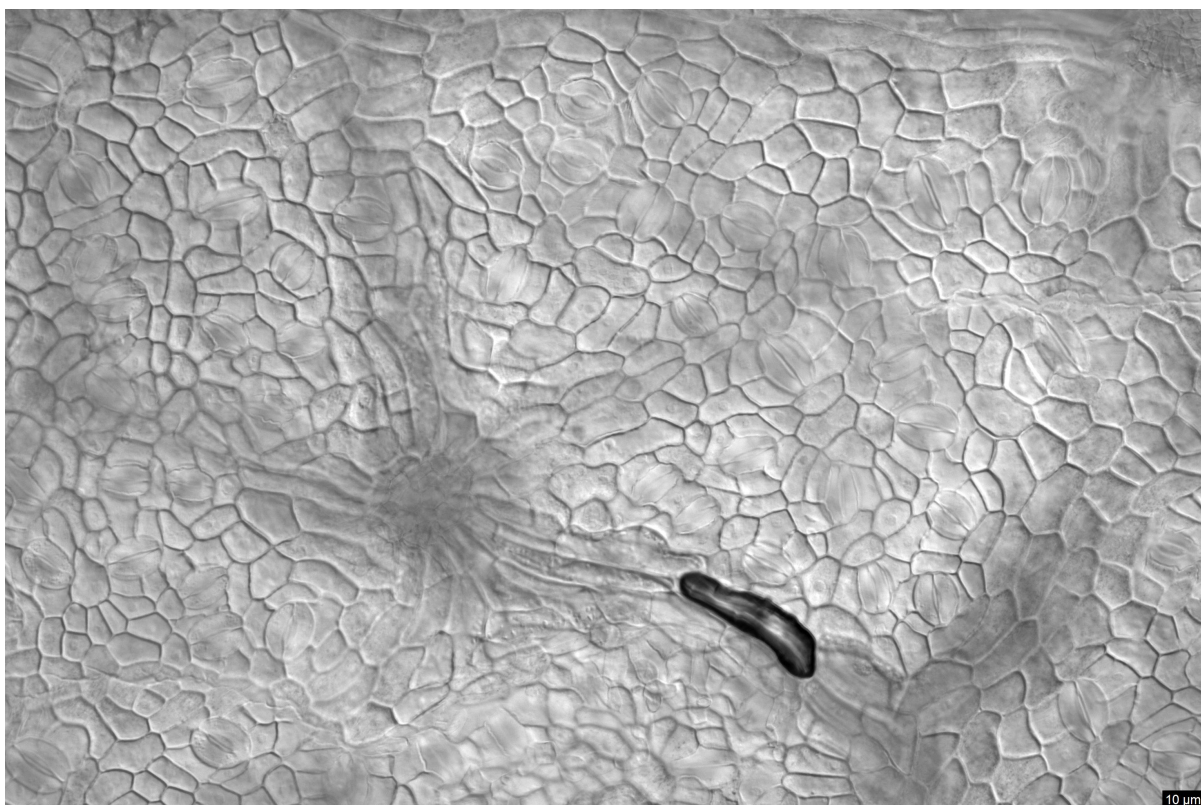


FIGURE 1. EXAMPLE IMAGE DEMONSTRATING STOMATA, TRICHOMES, VEINS, 'NOTHING' AND BLURRY AREAS

Results

Discussion

References

- [1] B. Dwyer, J. Nelson, T. Hansen, et al., *Roboflow*, version 1.0, 2024. [Online]. Available: <https://roboflow.com>.